Approximation by Simple Structures

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Smoothly Approximable Structures

Definition

A **smoothly approximable** structure is a countable ω-categorical structure $\mathcal{M}$ for which there is an ascending chain of finite homogeneous substructures $\mathcal{N}_i$ such that $M = \bigcup_{i<\omega} N_i$.

- Introduced by Lachlan, theory developed by Cherlin and Hrushovski
- Influential in the development of simple theories

NSOP_1

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Motivation

Build understanding of NSOP_1 as a generalization of simple by way of smooth approximability instead of forking independence.

$$\downarrow^f : \downarrow^K :: \text{smoothly approximable} : ?$$

1999 - Nick Granger's Thesis

Example

T_∞: the two-sorted theory of an infinite-dimensional vector space over an algebraically closed field with a non-degenerate, symmetric or alternating bilinear form.

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- Over a finite field, this is smoothly approximable
- Approximated using a sequence of finite-dimensional subspaces
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Limitation

The definition of $\downarrow^\Gamma$ is specific to vector spaces and doesn't make sense outside of this context.

2013 - Gwyneth Harrison-Shermoen's Thesis

- Developed a nice abstract framework for approximation independent of a specific example, defining $\downarrow^{\lim}$ as the limit of non-forking independence in a directed system of substructures
- Proved various properties of $\downarrow^{\lim}$ that follow from simplicity of the approximating structures
- Showed T_∞ (mostly) fits into this framework, with $a \downarrow^{\lim}_{C} B$ in T_∞ being equivalent to $a \downarrow^{\Gamma}_{\mathrm{acl}(C)} \mathrm{acl}(B)\,\mathrm{acl}(C)$

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Limitation

Theoretically applied much more broadly, but still just the one example.

Generalized Approximation Framework (H-S)

Let $\mathcal{H}$ be a directed system of substructures of a structure $\mathcal{M}$. We say that a property holds "for large enough $\mathcal{N}$" to mean that there is some $\mathcal{N} \in \mathcal{H}$ such that for all $\mathcal{N}' \in \mathcal{H}$ with $\mathcal{N} \subset \mathcal{N}'$, the property holds in $\mathcal{N}'$.
We write $\bar{A} = \mathrm{acl}^{\mathcal{M}}(A)$, where $\mathcal{M}$ is the approximated structure.

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Definition

- A formula is *eventually true (false)* in $\mathcal{H}$ if it is true (false) for large enough $\mathcal{N}$ in $\mathcal{H}$.
- A partial type $\pi(x, b)$ *eventually forks (eventually doesn't fork)* over C if for large enough $\mathcal{N}$ in $\mathcal{H}$, $\pi(x, b)$ forks (doesn't fork) over $C \cap N$.
- A type $p(x)$ *eventually forks* over C if it contains a finite subtype that eventually forks over C. It *eventually doesn't fork* over C if every finite subtype eventually doesn't fork.
- $A \downarrow^{\text{lim}}_{C} B$ if for each finite tuple $a \in A$ there is $\mathcal{N}$ large enough such that $a \downarrow^{\mathcal{N}}_{\bar{C} \cap N} \bar{B} \cap N$, where $\downarrow^{\mathcal{N}}$ is non-forking independence in $\mathcal{N}$.

Generalized Approximation Framework (H-S)

Given $\mathcal{M} \models T$ and $\mathcal{H}$ a directed system of substructures $\mathcal{N}$ of $\mathcal{M}$ we say $\mathcal{H}$ *approximates* $\mathcal{M}$ if conditions 1 through 5 hold. Conditions 6 and 7 are originally stated as optional.

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5. Strong finite character of $\not\perp^{\text{lim}}$: If $A \not\perp^{\text{lim}}_C B$, then there is $\varphi(x, b, c) \in \text{tp}^{\mathcal{M}}(A/\bar{B}\bar{C})$ which eventually forks over $\bar{C}$ in $\mathcal{H}$.

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Examples

We show there are actually several examples of NSOP_1 theories approximated by simple structures in the sense of Harrison-Shermoen, suggesting that approximability is a relatively common property in NSOP_1.

The following are all examples of NSOP_1 theories that can be approximated in this framework (more or less):

- T_∞
- T^*_{feq}
- ω-free PAC fields

Example

The generic theory of a parameterized equivalence relation. On an object sort O and a parameter sort P, the relation $E_x(y, z)$ is defined such that for each $p \in P$, $E_p(y, z)$ is an equivalence relation on O with infinitely many classes, all of which are infinite.

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- Approximating by restriction to finitely many parameters fails to satisfy convergence of truth value with the formula $(\forall x, y)(\exists p)\neg E_p(x, y)$.
- Approximate instead by substructures $\mathcal{M}_n$ where we restrict to at most n classes with respect to each parameter. This satisfies all conditions except automorphism invariance.

Automorphism Invariance vs Directed System

Automorphism invariance for T^*_{feq} is in tension with $\mathcal{H}$ being a directed system. In order to get automorphism invariance, we need to consider not just a single chain of substructures $\mathcal{M}_n$, but the collection of all substructures of $\mathcal{M}$ that are isomorphic to some $\mathcal{M}_n$.

However, this is no longer a directed system.

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Solution

Take $\mathcal{H}$ to be the automorphism invariant collection of substructures and define $\downarrow^{\text{lim}}$ on a filtration of $\mathcal{H}$ to a directed system that satisfies the remaining axioms. As long as $\downarrow^{\text{lim}}$ is invariant of the choice of filtration, we can get all the same conclusions approximating via the filtration.

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- Forking is trivial in a structure where we restrict to finitely many classes, so limit independence along any filtration is trivial.
- This fixes the issue with automorphism invariance, so T^*_{feq} is approximable by the chain of $\mathcal{M}_n$'s.
- It also means $\downarrow^{\text{lim}}$ is trivial in the approximated theory. Since $\downarrow^{\text{lim}}$ inherits base monotonicity from forking in the approximation, even in cases where it has more structure it will not recover $\downarrow^{K}$ in strictly NSOP_1 theories.

Non-Examples

There are structures with simple theories that are not smoothly approximable (e.g the random graph). It stands to reason the same should be true of the NSOP_1 analogue.

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We can find concrete examples of some non-approximable structures borrowing arguments from the pseudofinite context, via this fact about convergence of truth value.

Theorem

Suppose $\mathcal{H}$ is a directed system of substructures of $\mathcal{M}$ satisfying convergence of truth value. Then $\mathcal{M}$ is elementarily equivalent to an ultraproduct of elements of $\mathcal{H}$.

Strategy

The following lemma is adapted from an argument by Macpherson & Tent (2024) used to show certain structures are not pseudofinite.

Lemma

Let (V, W, β) be a 2-sorted structure where V, W are vector spaces over a field F (each viewed in the two-sorted language $\mathcal{L}_{vs}$ of a vector space over F) of dimension at least 2 and $\beta : V \times V \to W$ is bilinear. Let χ be a sentence expressing this information and ψ be the sentence

$$(\forall \text{ linearly independent } v_1, v_2 \in V)(\forall w_1, w_2 \in W)$$

$$(\exists z \in V)(\beta(v_1, z) = w_1 \wedge \beta(v_2, z) = w_2).$$

Then $\chi \wedge \psi$ has no model of finite Morley rank.

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Examples of Non-Approximable Theories

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- Vector spaces with generic k-linear forms
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Question

What about the generic binary function?

Recovering Kim Independence

Recall that $\downarrow^{\text{lim}}$ does not recover $\downarrow^{K}$ for strictly NSOP_1 theories, because of the discrepancy regarding base monotonicity.

Is there another means by which we can recover $\downarrow^{K}$ in the limit solely from properties of the approximation?

Recovering Kim Independence

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Is there another means by which we can recover $\downarrow^{K}$ in the limit solely from properties of the approximation?

Idea

Build a notion of limit *dimension*, constructed as the limit of dimension in the approximation, such that Kim-independence corresponds to a drop in limit dimension.

Motivation: Dobrowolski's Dimension for T_∞

Definition

Given a definable set X over a model $M = \bigcup N_r$ (an approximating sequence in the sense of Granger) let $\dim(X)$ be the function $f : \omega \to \mathbb{Z}$ given by $f(r) = \mathrm{rk}_{N_r}(X \cap N_r)$, where rk_{N_r} denotes Morley rank in N_r.

Dobrowolski showed that for any definable set X, $\dim(X)$ is almost everywhere equivalent to a linear function, which can then be ordered lexicographically as pairs (slope, intercept).

This notion of dimension does not recover $\downarrow^K$, but is the motivation behind our notion of limit dimension.

Limit Dimension

We would like to construct a notion of dimension constructed from the approximation that corresponds to $\perp^K$ in the limit. Dimension naturally gives rise to the following independence relation:

Definition

Let $\perp^{\dim}$ be the independence relation representing a drop in limit dimension:

$$A \perp^{\dim}_C B \iff \dim_C(A/C) = \dim_C(A/B)$$

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In strictly NSOP_1 theories, $\downarrow^K$ does not satisfy base monotonicity. This means our dimension will be defined over a base set C.

Limit Dimension

Definition

Let $\mathcal{M} \models T$ be a structure approximated by $\mathcal{H}$. For $\mathcal{N} \in \mathcal{H}$ large enough, a base set $C \subseteq M$, and a set X definable over M, let $\dim^{\mathcal{N}}_{C \cap N}(X \cap N)$ be a notion of dimension valued in a partially ordered abelian group G.
We call $\dim^{\mathcal{N}}_{C \cap N}(X \cap N)$ the **approximated dimension** of X over C in $\mathcal{N}$.
The **limit dimension** of X over C is the equivalence class $[f]$ of functions $\mathcal{H} \to G$ almost everywhere equivalent to the function $f(\mathcal{N}) = \dim^{\mathcal{N}}_{C \cap N}(X \cap N)$ for large enough $\mathcal{N}$.

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In T^*_{feq}, the approximated dimension is a tuple $(\text{rk}_=(X), (\text{rk}_p(O(X)), \deg_p(O(X)))_{p \in C})$ where $\text{rk}_=$ is Morley rank in $L = \{=\}$, rk_p is Morley rank in $\mathcal{N}_p$, and $\deg_p$ is the multiplicity.

Limit Dimension Axioms

Theorem

If for large enough $\mathcal{N}$ the approximated dimension satisfies invariance, algebraicity, union, fibration, additivity, and eventual uniform definability, then $\downarrow^{\dim}$ satisfies symmetry, monotonicity, strong finite character, and extension.

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- Does the notion of parameterized dimension for T^*_{feq} also recover Kim-independence when applied to other parameterized theories?
- Is there a notion of dimension for T_∞ fitting this axiomatic framework?